Andrey Sidorenko, PhD

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EXECUTIVE SUMMARY

AI/ML engineer and research leader with 15+ years bridging deep science and production AI. Track record of designing, training, and deploying generative AI models (diffusion models, auto-regressive networks, transformers, etc.) at enterprise scale with formal privacy guarantees. Experienced in end-to-end AI/ML lifecycle: data engineering, model architecture design, distributed training, inference optimization, MLOps, and analytics. Led teams of engineers/researchers; translated research into revenue-generating products.

PROFESSIONAL EXPERIENCE

• Head of AI Research | MOSTLY AI, Austria | Oct 2021–Apr 2026

- Lead the company's AI research strategy and execution, focusing on high-fidelity, privacy-preserving synthetic data generation for enterprise clients in regulated industries.
- Drive the vision and implementation of next-generation generative AI models, advancing the boundaries of realism, scalability, and data compliance.
- Launched and directed LISA (Language + Images + Structured data + Audio) and DARTS (Domain Adaptation & Rule-based Transformation & Simulation) projects, pioneering multimodal and rule-driven data simulation AI frameworks.
- Partner with product and sales teams to integrate AI innovations into customer-facing solutions, increasing client adoption and positioning MOSTLY AI as a global leader in synthetic data.
- Lead the team of ML/AI researchers.
- Represented company AI research at international conferences.

• Software Development Team Lead | MOSTLY AI, Austria | Mar 2019–Oct 2021

- Led the development of the AI Engine powering the company's synthetic data platform.
- Managed a team of developers to architect distributed systems for large-scale synthetic data processing, ensuring quality, privacy, and computational efficiency.
- Collaborated closely with data scientists and business stakeholders to align technical innovations with evolving customer needs.

• Researcher | Technische Universität Wien, Austria | Feb 2010–Mar 2019

- Built the Vienna Microkelvin Laboratory (WIENμK) - the first microkelvin laboratory in Austria.
- Led projects exploring quantum-critical materials and engineered complex real-time data acquisition systems for a sophisticated scientific facility, worth €1 million, saving numerous staff hours.
- Enhanced the FFT noise analysis process through GPU parallelization, reducing over 1,000 CPU hours and increasing production speed by 200%.

• Researcher | Universita di Parma, Italy | Feb 2002–Jan 2010

- Led projects focusing on characterization of spintronics materials be means of NMR.
- Initiated and led a transition project from Matlab to Python, resulting in an annual saving of €20,000 for the institute by cutting license fees by half.
- Developed software to analyze experimental data resulting in designing the first organic spintronic devices and various innovative tools.

EDUCATION

• PhD in Physics and Mathematics | Moscow State University, Russia | Oct 1997–Apr 2000

• M.S. in Physics | Bryansk State University, Russia | Jul 1992–May 1997

CORE COMPETENCIES AND TECHNOLOGY STACK

AI / ML & Data Science	Engineering & Infrastructure	Leadership
• LLMs, Transformers, Diffusion models, VAEs, GANs, Auto-regressive generative networks • Fine-tuning (LoRA, QLoRA, PEFT) & RLHF; Prompt Engineering • Agentic AI, LangChain, LangGraph, RAG pipelines • Synthetic data, differential privacy, federated learning, anonymization • Multimodal generative models • NLP, embeddings, semantic search, vector DBs • Statistical modeling, Bayesian inference, causal analysis • Analytics & Visualization	• Python (expert), JavaScript/TypeScript, Fortran, SQL, Bash • Multi-GPU distributed training (DeepSpeed, FSDP) • Inference optimization (TensorRT-LLM, vLLM, ExLlama v2, ONNX Runtime, DeepSpeed-MII, Optimum-NVIDIA) • Cloud: AWS (EC2, S3, Lambda), GCP (Vertex AI), Microsoft Azure (Foundry), Runpod • MLOps: Docker, Kubernetes, CI/CD • ML/DL Frameworks (PyTorch, TensorFlow, FLAX/JAX, scikit-learn, XGBoost, LightGBM, HF, etc.) • Data Engineering: Pandas, NumPy, SciPy, PySpark • Databases: PostgreSQL, MongoDB, FAISS • Git/GitHub, Linux, Bash • Vibe coding: Codex, Claude Code, Antigravity	• AI strategy, research roadmap • Cross-functional team leadership (R&D, Product) • Mentoring ML researchers & engineers • Public speaking, thought leadership • Enterprise consulting and entrepreneurship

DELIVERED AI/ML PROJECTS

• TabularARGN — Production Generative Model for Synthetic Data | MOSTLY AI, Austria | 2022–2026

Challenge: Enterprise clients in regulated industries needed high-fidelity synthetic data preserving complex statistical dependencies while guaranteeing formal privacy (GDPR/HIPAA).

Solution: Designed TabularARGN: a discretization-based auto-regressive neural network for tabular data. Integrated differential privacy with novel loss-clipping in training loops. Built multi-GPU distributed training pipeline. Implemented comprehensive privacy evaluation via systematic membership-inference attacks.

Outcome: SOTA results in fidelity, ML utility, and privacy; 10–100× faster than competitors. Became the core engine of MOSTLY AI's platform. Open-sourced and published: arXiv 2501.12012, 2508.06647.

• LLM-Based Tabular Generation via Probability-Driven Prompting | MOSTLY AI, Austria | 2024–2025

Challenge: LLM-based synthetic data methods failed to preserve categorical feature dependencies due to sequential auto-regressive generation limitations.

Solution: Introduced probability-driven prompting: LLMs estimate conditional distributions instead of generating values directly. No fine-tuning required.

Outcome: Superior statistical fidelity over existing LLM tabular generation. Published: arXiv 2505.02659.

• LISA & DARTS — Multimodal & Rule-Driven Simulation Framework | MOSTLY AI, Austria | 2024–2026

Challenge: Clients required synthetic data spanning text, images, structured data, and audio with domain-specific rules (temporal priors, regulatory constraints) beyond purely neural approaches.

Solution: Built LISA (Language + Images + Structured + Audio) for multimodal generation and DARTS for domain-adaptive rule-based transformations. Integrated calendar features, interventions, and external priors into sequence models.

Outcome: Positioned MOSTLY AI as global leader in multimodal synthetic data.

• Synthetic Data QA Framework (Open-Source) | MOSTLY AI, Austria | 2024–2025

Challenge: No standardized methodology existed for quantitative tabular synthetic data evaluation across fidelity, utility, and privacy dimensions.

Solution: Designed holdout-based benchmarking with distributional comparisons, embedding similarity, nearest-neighbor distances, and privacy metrics. Released as open-source Python package.

Outcome: Industry-standard evaluation tool. Published: arXiv 2504.01908.

• AI Engine — Distributed Synthetic Data Platform | MOSTLY AI, Austria | 2019–2021

Challenge: Needed scalable production AI engine handling large-scale data processing with strict quality, privacy, and efficiency requirements.

Solution: Led team to architect distributed AI engine: data ingestion, training pipelines, model serving, and analytics dashboard for comparing synthetic vs. original data distributions.

Outcome: Core commercial platform powering enterprise synthetic data generation at scale.

• High-Performance Inference Optimization Pipeline | MOSTLY AI, Austria | 2023–2026

Challenge: LLM-based generation and fine-tuned models required fast, cost-efficient inference for production workloads across tabular and text domains.

Solution: Designed and benchmarked inference pipelines using TensorRT-LLM, DeepSpeed-MII, vLLM, ExLlama v2, and Optimum-NVIDIA. Optimized batch strategies, quantization (GPTQ, AWQ, bitsandbytes), and KV-cache management.

Outcome: Reduced inference latency by 5–10× and cost by 60%+ across workloads.

SELECTED AI/ML PUBLICATIONS

- A. Sidorenko *et al.* “TabularARGN: An Auto-Regressive Generative Network for Tabular Data Generation.” EurIPS'25 Workshop on AI for Tabular Data.
- A. Sidorenko, P. Tiwald. “Privacy-Preserving Tabular Synthetic Data Generation Using TabularARGN.” arXiv: 2508.06647, 2025.
- A. Sidorenko *et al.* “Benchmarking Synthetic Tabular Data: A Multi-Dimensional Evaluation Framework.” arXiv: 2504.01908, 2025.
- A. Sidorenko. “A Note on Statistically Accurate Tabular Data Generation Using Large Language Models.” arXiv: 2505.02659, 2025.
- I. Krchova *et al.* “Democratizing Tabular Data Access with an Open-Source Synthetic-Data SDK.” arXiv: 2508.00718, 2025.

Physics: 50+ papers incl. Science Advances, Nature Physics, PRL (1,500+ citations). Full list on Google Scholar.